SUMMER INTERNSHIP REPORT
On
GENERATIVE AI & MACHINE LEARNING

Undertaken at

Coding Blocks School of Technology

Submitted in partial fulfillment of the requirements for the degree of

Bachelor of Technology in Artificial Intelligence and Machine Learning

I would like to express my deepest appreciation to Coding Blocks School of Technology for providing me with the opportunity to complete my summer internship as an AI Engineer. The hands-on experience and mentorship I received were invaluable in shaping my understanding of Generative AI and Machine Learning.

I also want to extend my gratitude to the faculty of IILM University for equipping me with the foundational knowledge necessary to succeed in this role and for their continuous guidance throughout my academic journey.

EXECUTIVE SUMMARY

This report documents the work, learnings, and projects completed during my 6-week summer internship at Coding Blocks School of Technology from 1st June 2026 to 14th July 2026. Working as an AI Engineer, the primary objective of this internship was to bridge the gap between theoretical data science concepts and practical, real-world application development.

During the internship, I focused heavily on Generative AI, Large Language Models (LLMs), Prompt Engineering, and Machine Learning. I successfully developed and deployed three major projects: an intelligent AI Agent (QI Agent), an automated AI Research Paper Intelligence System, and a predictive Telco Customer Churn Analysis model. This experience significantly enhanced my technical proficiency in Python, NLP, and agentic workflows.

COMPANY PROFILE

Coding Blocks School of Technology is a premier educational institute and technology hub renowned for its focus on software development and cutting-edge tech education. The organization specializes in teaching and building advanced systems in Artificial Intelligence, Machine Learning, Data Science, and Web Development. By providing an immersive, hands-on environment, Coding Blocks

equips developers and engineers with the skills required to build scalable projects and solve complex real-world industry problems.

INTERNSHIP OBJECTIVES

The core objectives set for this 6-week internship included:

Hands-on AI Experience: To gain practical experience in building, fine-tuning, and deploying modern Artificial Intelligence models.

Generative AI Mastery: To understand and implement advanced GenAI concepts, including LLM integrations, prompt engineering, and the creation of agentic workflows.

Machine Learning Application: To apply foundational Machine Learning techniques and Exploratory Data Analysis (EDA) to real-world datasets for robust predictive analysis.

End-to-End Development: To experience the complete software development lifecycle for AI applications, from initial architecture ideation to final execution.

PROJECT DETAILS & ARCHITECTURE

During the internship, I operated as an AI Engineer and was responsible for developing the following three core projects:

Project 1: QI Agent

Repository: github.com/Requireding/QI_AGENT

Description: I developed an intelligent AI agent system designed to handle complex, automated tasks and smart query resolution.

Technologies Used: Generative AI, Prompt Engineering, Agentic Workflows, Python.

Impact & Results: The project successfully streamlined query handling by creating a dynamic AI assistant that accurately adapts to varying user inputs and executes multi-step logical operations without human intervention.

Project 2: AI Research Paper Intelligence System

Repository: github.com/Requireding/AI-Research-Paper-Intelligence-System

Description: I built a Generative AI application designed to read, analyze, summarize, and extract key insights from complex academic research papers automatically.

Technologies Used: Large Language Models (LLMs), Natural Language Processing (NLP), Document Parsing Frameworks.

Impact & Results: The system significantly reduced the time required for academic literature reviews by automating the extraction of core methodologies, data points, and research conclusions from large PDF documents.

Project 3: Telco Customer Churn Analysis

Repository: github.com/Requireding/telco-customer-churn-analysis

Description: I conducted comprehensive data analysis and predictive modeling on telecommunications data to forecast which customers were highly likely to cancel their service.

Technologies Used: Machine Learning algorithms, Exploratory Data Analysis (EDA), Predictive Modeling, Python (Pandas, Scikit-Learn).

Impact & Results: The model successfully identified key statistical metrics and user behaviors that lead to customer churn, enabling potential business stakeholders to implement targeted, data-driven retention strategies.

KEY LEARNINGS & SKILLS ACQUIRED

Generative AI & LLMs: Deepened my understanding of how to programmatically interact with Large Language Models, optimize prompts for accurate outputs, and build autonomous agents.

Data Pipelines & Processing: Gained extensive experience in cleaning, structuring, and analyzing raw datasets to extract actionable business insights.

Problem-Solving & Architecture: Improved my critical debugging skills and learned how to properly plan system architecture before writing code.

Version Control: Enhanced my professional proficiency in managing Git repositories, version control, and writing clear documentation on GitHub.

CONCLUSION

The 6-week internship at Coding Blocks School of Technology was a highly rewarding and transformative experience. It provided me with a platform to apply the theoretical knowledge gained at IILM University to practical, industry-standard AI engineering tasks. By successfully completing projects in Generative AI and Machine Learning, I have developed a robust skill set that prepares me for future challenges in the rapidly evolving field of Artificial Intelligence.